



LTE Based Satellite Networks

Abstract:

Traditional solutions using Very Small Aperture Terminal (VSAT) networks as satellite backhaul involve a high cost of development with low flexibility in service deployments. Companies have increasingly preferred to use customized 3GPP networks directly over satellite for better flexibility. Recent trends show a high number of Low Earth Orbit (LEO) satellites being launched, removing the performance barriers of high delays and jitter on Geostationary (GEO) satellites. This is encouraging the deployment of 3GPP 4G/LTE networks directly over satellite links.

Author:

Amit Maheshbhai Bhatt, Chief Architect – Product Engineering,
Sasken Technologies Ltd.



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Preface

The number of connected objects and devices on Earth is now greater than the world's population. However, there are still some geographical areas where coverage of terrestrial networks is unavailable. These are remote regions of the world where the population is so low that mobile operators cannot economically deploy network infrastructure. These include areas where the environment is so hostile that it is almost impossible to deploy infrastructure — such as open oceans, mountains, deserts, and forest regions. Satellite communication is the only viable solution to provide coverage and connectivity to these regions.

There has been a continuing desire to complement mobile terrestrial networks with satellite networks for several decades. Developing these solutions has presented difficulties in forms of the viability of business models and technical challenges.



Traditional solutions

Much progress has been made since adaptations of 2G cellular systems for satellite communications started in the late 1990s. But the solutions developed so far have relied on proprietary adaptations of networks and devices and have not been able to benefit from economies of scale in the mass market.

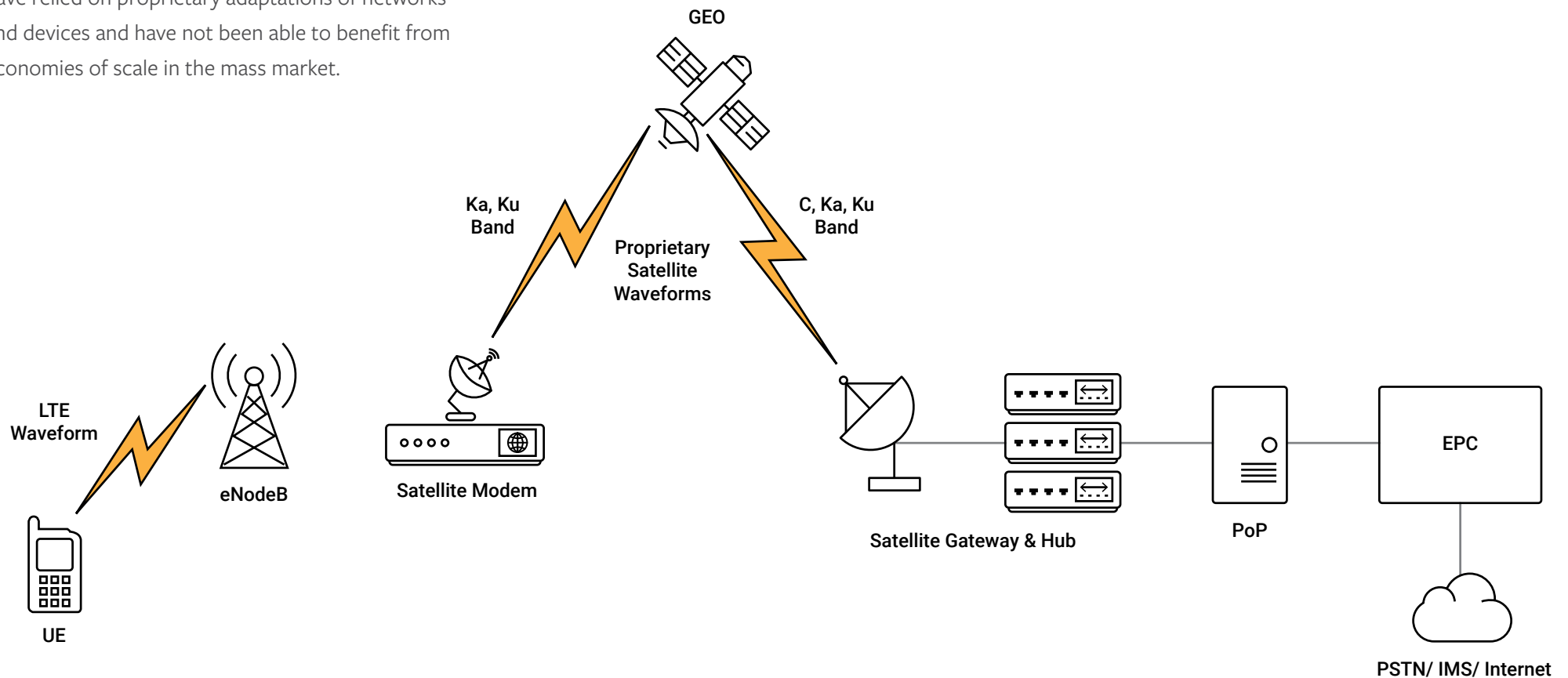


Figure 1 Satellite Backhaul for Terrestrial Networks



Classical communication satellites are either broadcasting or Internet relay satellites, which have been mainly designed to provide direct real-time connectivity between different ground locations. Direct relay connection requires simultaneous connection to both the satellite and terrestrial transceiver chains.

The traditional satellite communication network has proprietary interfaces and standards for networks, terminal elements, network slicing, QoS, resource management, service prioritization, etc. The 3GPP core network integrates with the satellite network via POP (Point of Presence), which interworks interfaces and services in the control and user planes.

The satellite network provides a tunnel to a network slice between EPC and eNodeB via the satellite gateway, hub, and modem. The eNodeB further uses LTE interfaces and waveforms for UEs at remote deployments. EPC, eNodeB, and UE protocols (timers, ARQ, timing advance, retransmissions, etc.) are tuned for satellite delays, jitters, and QoS aspects.

With proprietary solutions on satellite networks, the development, maintenance, and deployment costs are high. This results in very high subscription rates as compared to the same set of services on standards-based terrestrial solutions. Solution upgrades for technological advancements, scale, and performance are very expensive for proprietary systems, and time to market is longer.

Above factors make traditional satellite-based solutions affordable only to premium businesses and government funded deployments. This results in reduced popularity and access to the mass market.



Adapted solutions using terrestrial technologies

Initiatives have been on to standardize satellite communications and integrate them with terrestrial systems for a long time. But this has not yielded results in 3GPP until 5G standardization, mostly because the interests of terrestrial mobile operators have been prioritized.

This has resulted in a few independent and proprietary adaptations of terrestrial standards, initiated, and used by several satellite companies. This includes the GMR-1 series from ETSI/TIA, the GMR2/2+ from Lockheed Martin and Inmarsat, and the GMR2-4G from Lockheed Martin, among others. With increasing hunger for satellite-based connectivity in segments such as industrial IoT and luxury cruises, several consortiums have moved ahead with the deployment of satellite systems derived from terrestrial LTE. Depending on the use cases, the satellite constellations are on Geostationary (GEO), Medium Earth Orbit (MEO), or Low Earth Orbit (LEO) orbits. Independent of orbital configurations, the overview of these systems is as follows:

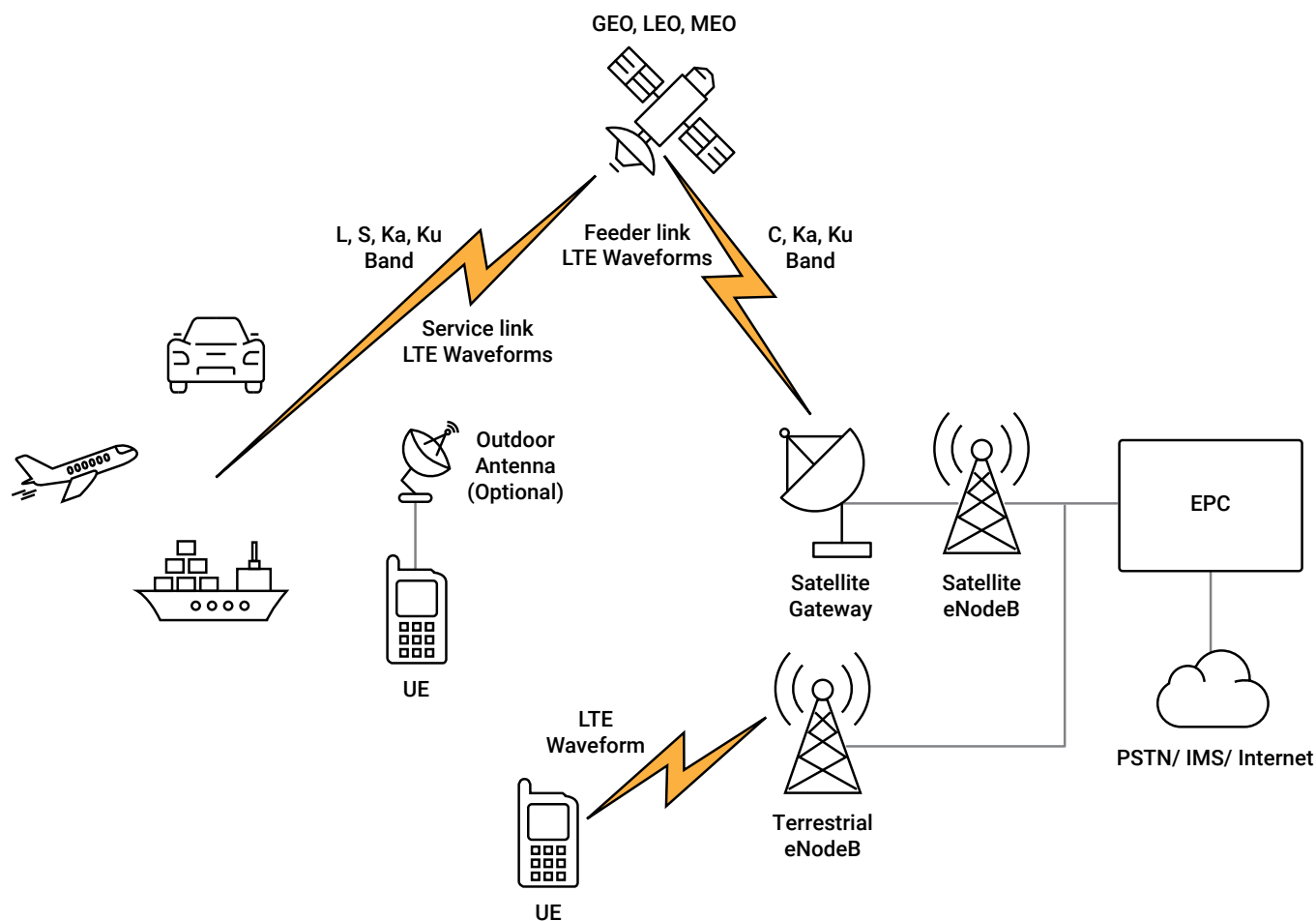


Figure 2 LTE Over Satellite (Transparent Mode)



Adaptation challenges

In this approach, the LTE air interface is modified for performance in the satellite environment. The satellite UEs and satellite eNodeBs are modified versions of their corresponding terrestrial products. The satellite acts as a transparent reflector between the service and feeder links. The core remains an unmodified terrestrial EPC. While UEs and eNodeB are derivative products, there is a significant reduction in cost as the terrestrial LTE ecosystem provides the satellite vendor with a choice of suppliers for the base products. Commercial networks such as OneWeb, AST SpaceMobile, and several critical communication networks around the world are in ongoing trials using LTE-based satellite networks.

The challenges in adapting a terrestrial standard for satellite environment are summarized below.

Satellite orbital mobility

Satellites in MEO/LEO orbits move fast resulting in satellite visibility only for a limited period from any location on earth. Multiple satellites are launched to provide continuous data connectivity. This results in periodic handovers from one satellite (eNodeB) to another satellite (eNodeB) as satellite visibility changes. This also results in two or more antenna, along with an associated receiver and optionally a transmitter system, at terminals to provide uninterrupted data connectivity.

Satellite delay

Satellites are far higher in elevation than terrestrial base stations, with GEO being almost 36,000 km away. This results in much longer round-trip time (RTT), requiring all closed loop procedures as defined in terrestrial specifications to be modified.

Satellite channels link budget

Satellites are far away, necessitating higher transmit power, lower receiver sensitivity, customized outer and inner loop power control, customized error rates, and ACM methods, etc. This results in defining new power classes and other radio capabilities for devices.



Cell and beam topology

Satellite beams are much larger in size as compared to terrestrial cells, resulting in adaptations needed for near-far problems, handover, cell reselection, and timing advance.

Satellite wireless channel

Satellite communication works online-of-sight principles, passing through atmosphere and environment that terrestrial signals do not experience. Satellite channels do not have the benefits of multipath environments and see very high doppler effects, especially on MEO/LEO satellites. This is also the case for fast-moving terminals like airplanes and drones.

Network slicing and Quality of Service

Concepts of terrestrial network slicing and quality of service can be directly deployed in the satellite system. This makes developing systems for service management, subscriber management, billing, and resource control easier. Increased RTT and reduced receiver sensitivity necessitate the definition of new services and QoS classes.

Time To Market

LTE-based approaches allow for the use of COTS and ready-to-deploy components for CN, RAN, and terminals with nominal customization. This reduces the time required to develop and upgrade solutions. Terrestrial eco system for overall network management enables deployment and operation of the network easier than proprietary systems. Subscriber management, O&M control, device and network commissioning and upgrades, etc. are examples.

The reduced cost of development and upgrades and the early time-to-market result in a reduction of the subscription cost for the services offered by the network.



3GPP standardization

With increasing demand for support of satellite-based data communications, 3GPP has included support for satellite communication in its Release 17 set of specifications. Based on studies carried out in Release 15 and Release 16, the recently frozen Release 17 has included Non-Terrestrial Networks (NTN) for 5G NR. NTN includes satellite communication using a transparent payload mechanism based on 5G NR in Release 17 and has also been extended to NB-IoT and eMTC variants of LTE.

A satellite standard closely resembling terrestrial standards will encourage the terrestrial component supplier ecosystem to bring out mass-market components for the satellite segment. This will drive service costs down and encourage larger-scale deployments.

Conclusion

The traditional solution of using VSAT networks as satellite backhaul is inflexible for terrestrial network planning and service deployments. This motivated the proprietary adaptation of cellular standards for the satellite environment. Increasing demand for satellite connectivity has encouraged several consortiums and countries to adapt the LTE standard for satellite systems, targeting commercial as well as critical communication. The terrestrial LTE component ecosystem has helped drive down the cost of deployment and upgrades. Availability of mass market satellite communication components is a likely prospect, with support for satellite coming into 3GPP standardization from Release 17. This will bring further economies of scale for satellite communication networks, and the implicit reduction in service costs will encourage larger-scale deployments.

With 20+ years of engineering excellence in satcom products, Sasken's portfolio spans access network SW, device SW, and complete device development. This includes proprietary technologies and adaptations of 3GPP technologies. Handsets, modules, and ruggedized devices completely designed by Sasken remain deployed worldwide. Sasken engineers are engaged in commercialization efforts for two LTE-based satellite communication systems, with scope ranging from devices to eNodeB SW. Complemented by a 30-year track record in terrestrial ETSI/3GPP based RAN and UE SW, Sasken is collaborating with customers to accelerate 3GPP NTN deployments.



About the Author

Amit Maheshbhai Bhatt has been with Sasken for more than 23 years, spearheading innovation and setting engineering standards for Sasken's product engineering business. Starting with a path-breaking 3G proof-of-concept in early 1999, Amit anchored the development and worldwide commercialization of Sasken's 3G Protocol Stack. This laid the foundation for the billion+ devices enabled by Sasken protocol stacks.

Amit subsequently led the development of GMR-2P satellite phones and modules designed by Sasken for Inmarsat. This involved full device development, productization and manufacturing support. These satellite products have been deployed in leading markets worldwide.

Together with mentoring Sasken teams involved in the engineering of LTE-based satellite devices, Amit is anchoring key programs in newer areas. This includes the development of devices and network SW for a VSAT based system having multi-orbit constellations.

About Sasken

Sasken is a specialist in Product Engineering and Digital Transformation providing concept-to market, chip-to-cognition R&D services to global leaders in Semiconductor, Automotive, Industrials, Consumer Electronics, Enterprise Devices, SatCom, Telecom, and Transportation industries. For over 32 years and with multiple patents, Sasken has transformed the businesses of 100+ Fortune 500 companies, powering more than a billion devices through its services and IP.





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marketing@sasken.com | www.sasken.com

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